

# Methods: frequency-resolved active-space downfolding of the electron–defect vertex and the defect spectral function

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We describe a first-principles route from a defect supercell to the electron–defect spectral function  $A(\mathbf{k}, \omega)$  of an isolated defect on an arbitrarily fine  $k$ -mesh. The scattering vertex is downfolded onto a small active manifold by an exact Feshbach partition, with the rest-space contribution resummed to all orders and resolved in frequency by a block Lanczos continued fraction. The dressed vertex is carried to fine  $k$  by a pair-Wannier transform, and the  $T$ -matrix is closed on a real-space cluster whose dimension is independent of the fine mesh. Each step is gated by a quantitative test: an operator-level identity, level-by-level comparison with a supercell reference at two supercell  $k$ -points, a leave-one-out test of the interpolation, and closure of the end-to-end chain onto the directly diagonalized quasiparticle levels to 0.4 meV.

## I. OVERVIEW

The object of interest is the electron–defect self-energy of a dilute, randomly distributed defect,

$$\Sigma_{mn}^{\text{ed}}(\mathbf{k}, \omega) = n_d T_{mn}(\mathbf{k}, \mathbf{k}; \omega), \quad (1)$$

and the spectral function that follows from it,

$$A(\mathbf{k}, \omega) = -\frac{1}{\pi} \text{Im Tr}[\omega + i\eta - H_0(\mathbf{k}) - \Sigma^{\text{ed}}(\mathbf{k}, \omega)]^{-1}. \quad (2)$$

Three obstacles stand between a supercell calculation and Eqs. (1)–(2). First, the scattering must be summed to all orders, not truncated at the Born level, because a defect that binds a state is not perturbative. Second, the sum must be resolved in frequency: freezing it at a reference energy is exact only there. Third, a supercell fixes both the defect concentration and the  $k$ -mesh, whereas Eq. (1) requires the isolated limit and a mesh dense enough to resolve  $A(\mathbf{k}, \omega)$ .

The method addresses the three in sequence. Section II splits the Hilbert space and eliminates everything outside a small active manifold exactly. Section III constructs the bare vertex on the primitive grid and states the conditions under which that construction is exact. Section IV resums the eliminated space to all orders and to arbitrary  $\omega$ . Section V carries the result to fine  $k$ , and Sec. VI closes the  $T$ -matrix in real space, where the concentration becomes a free parameter decoupled from the mesh. Section VII collects the tests.

Throughout, the illustrative system is a monolayer MoS<sub>2</sub> with a relaxed sulfur vacancy (V<sub>S</sub>), and, where stated, the isovalent substitutions O<sub>S</sub> and Se<sub>S</sub>.

## II. ACTIVE/REST PARTITION AND THE EXACT DOWNFOLDING IDENTITY

Let  $P_A$  project onto an active manifold  $A$  —  $n_b$  Bloch bands of the pristine crystal at each of  $N_k$  points of a coarse mesh, so  $N_A = n_b N_k$  — and let  $P_R = 1 - P_A$ .

With  $H = H_0 + \Delta V$ , where  $H_0$  is the pristine Kohn–Sham Hamiltonian and  $\Delta V$  the defect potential, Feshbach–Löwdin partitioning of  $(\omega - H)^{-1}$  gives the projected Green function on  $A$  *exactly*,

$$G_A(\omega) = [\omega - \varepsilon_A - \tilde{V}(\omega)]^{-1}, \quad (3)$$

with the dressed vertex

$$\tilde{V}(\omega) = \frac{1}{N_k} [M_{AA} + \Sigma^R(\omega)], \quad M_{ab} = \langle \psi_a | \Delta V | \psi_b \rangle, \quad (4)$$

and the rest self-energy

$$\Sigma^R(\omega) = V_{AR} [\omega - P_R H P_R]^{-1} V_{RA}. \quad (5)$$

Equation (5) contains *every* path that leaves  $A$ , scatters an arbitrary number of times within  $R$ , and returns — all orders in  $\Delta V$  restricted to  $R$ , zeroth order in  $A$ . The complementary paths, those that re-enter  $A$ , are generated by inverting Eq. (3) or, equivalently, by the ladder of Sec. VI. Because  $P_A P_R = 0$ , no path is counted twice; this is the structural reason a  $T$ -matrix built on  $\tilde{V}$  with an active-space propagator requires no subtraction.

Two consequences are used later. (i) Since  $\Sigma^R$  depends on  $\omega$ , the spectral weight carried by  $A$  is less than unity, with the quasiparticle renormalization  $Z = [1 - \partial_\omega \tilde{V}]^{-1}$ ; this must be reported when  $A$ -space spectral weights are compared with total ones. (ii) The factor  $1/N_k$  in Eq. (4) is the Born–von–Kármán normalization: a calculation on an  $N_k$ -point mesh describes one defect per  $N_k$  primitive cells.  $N_k$  is therefore not only a sampling parameter but also the dilution, a point we return to in Sec. III C.

## III. THE DEFECT VERTEX ON THE PRIMITIVE GRID

### A. Folding the supercell potential

$\Delta V$  is obtained as the difference of the self-consistent local potentials of a defect supercell and the corresponding pristine supercell, aligned in the vacuum. It is represented on the supercell real-space grid. The matrix

element between primitive-cell Bloch states requires it folded onto the primitive grid with the exact momentum transfer,

$$V_{\mathbf{q}}(\mathbf{r}) = \sum_{\mathbf{R} \in \text{cell}} \Delta V(\mathbf{r} + \mathbf{R}) e^{i\mathbf{q} \cdot (\mathbf{r} + \mathbf{R})}, \quad \mathbf{q} = \mathbf{k}' - \mathbf{k}, \quad (6)$$

where  $\mathbf{R}$  runs over the primitive lattice vectors contained in the supercell box. Equation (6) is evaluated on the fly for each pair, with the phase running over the *whole* box and  $\mathbf{q}$  taken exactly; wrapping  $\mathbf{q}$  into the first Brillouin zone without the compensating phase would introduce a spurious factor. The implementation follows the convention of Ref. 9.

### B. Commensurability of the fold

The sum in Eq. (6) is realized as a real-space modulo map,  $r_i^{\text{cell}} \rightarrow r_i^{\text{cell}} \bmod n_i^{\text{prim}}$ . This is exact only if

$$n_i^{\text{cell}} = N_i^{\text{sc}} n_i^{\text{prim}} \quad \text{on every axis}, \quad (7)$$

i.e. the ratio must *be* the supercell multiplicity, not merely an integer. The failure is silent: in-gap states still appear at roughly the right energies while  $C_3$ -degenerate doublets split by tens of meV and  $\|\hat{V}\|$  changes by tens of percent. A merely integral ratio fails just as badly — a 240-point cell of a  $6 \times$  supercell folded onto a 30-point primitive grid (ratio 8) returned 15 states in a window that holds 5. Degeneracy splitting of a symmetric defect is the cheap fingerprint; on commensurate grids the  $C_3$  pairs here are degenerate to  $0\text{--}4 \mu\text{eV}$ . The code prints both grids and the detected multiplicity and aborts when Eq. (7) fails.

Equation (7) can be satisfied from either side. Rather than refine the primitive grid to match a fine potential grid, we band-limit the potential onto a coarser commensurate grid by truncation in  $\mathbf{G}$ -space, which is exact for every Fourier component the coarse grid supports. For the case studied,  $240 \times 240 \times 300 \rightarrow 162 \times 162 \times 216$  discards  $4 \times 10^{-8}$  of the spectral weight of  $\Delta V$  and pairs with the natural  $27 \times 27 \times 216$  primitive grid; the cost of Eq. (6) falls in proportion to the grid, and quasiparticle levels move by  $\leq 0.1 \text{ meV}$ .

### C. The $k$ -mesh is free; $N_k$ is the dilution

The supercell is a container that captures one isolated defect, not a periodicity we wish to impose. Because the phase in Eq. (6) runs over the whole box,  $V_{\mathbf{q}}$  is the finite-box transform of a *localized*  $\Delta V$  and is defined, and generically nonzero, for arbitrary  $\mathbf{q}$ . The  $k$ -mesh therefore need not match the supercell.

The distinction is quantitative. Treating the box as a periodicity would set  $\langle \psi_{\mathbf{k}} | \Delta V | \psi_{\mathbf{k}'} \rangle = 0$  unless  $\mathbf{k} - \mathbf{k}' \in \{\mathbf{G}_{\text{sc}}\}$ . Measured on a  $12 \times 12$  mesh against a  $6 \times 6$

box, matrix elements between  $k$ -points differing by a non-supercell vector reach 0.975 of the largest within-supercell one: the periodic reading would discard most of the scattering channels.

Two inequivalent calculations therefore share the name “ $12 \times 12$ ”:

- (a) *Coset-decomposed.* Since  $\Delta V$  of a strictly periodic array cannot connect them, the mesh splits into  $(N_k/N^{\text{sc}})^2$  cosets of the supercell reciprocal lattice, each an independent run of the designed size, and each folding to one supercell  $k$ -point. The dilution stays at  $1/N^{\text{sc}}$ , which is what makes it comparable to a supercell calculation. This is a validation device (Sec. VII B), not the physics target.
- (b) *Direct.* One run on the full mesh: one defect per  $N_k$  cells, with  $\mathbf{q}$  resolved throughout the zone. This is the dilute limit the method targets.

The two agree where they must: the coset-diagonal blocks of the direct  $144\text{-}k$  vertex reproduce the four shifted  $36\text{-}k$  runs to  $2 \times 10^{-6}$  on a block scale of 66, compared gauge-invariantly through block eigenvalues.

## IV. THE REST SELF-ENERGY BY BLOCK LANCZOS

This section records the algorithm in full. An *active label* is a pair  $a = (n, \mathbf{k}_i)$  of band and coarse momentum,  $N_A$  their number; a *bundle*  $|\chi\rangle$  is one algebraic vector with a plane-wave component at *every* coarse channel,  $\chi_{\mathbf{k}'}(\mathbf{G}, \sigma)$ , of total length  $N_{\text{pw}}^\sigma N_k$  ( $\sigma$ : spinor component;  $N_{\text{pw}}^\sigma = 2N_{\text{pw}}$  for  $n_{\text{pol}} = 2$ ). The bundle inner product sums over channels,  $\langle \chi | \xi \rangle = \sum_{\mathbf{k}' \mathbf{G} \sigma} \chi_{\mathbf{k}'}^*(\mathbf{G}, \sigma) \xi_{\mathbf{k}'}(\mathbf{G}, \sigma)$ ; every orthogonality statement below refers to it.

### A. Source block: construction and projection

The resolvent (5) is needed only between the  $N_A$  sources

$$|\chi_a\rangle = P_R \Delta V |\psi_a\rangle, \quad P_R = 1 - P_A, \quad P_A = \sum_{b \in A} |\psi_b\rangle \langle \psi_b|. \quad (8)$$

Three details fix the implementation. (i)  $\Delta V |\psi_a\rangle$  uses the exact- $q$  fold of Sec. ??: the state resident at channel  $\mathbf{k}_i$  scatters into *every* channel  $\mathbf{k}'$  with the folded potential  $V_{\mathbf{k}_i - \mathbf{k}'}(\mathbf{r})$  of Eq. (??); no momentum is wrapped to a supercell zone. The nonlocal (Kleinman–Bylander) difference is added per output channel,  $\sum_{\mu\nu} |\beta_\mu^{\mathbf{k}'}\rangle D_{\mu\nu} \langle \beta_\nu^{\mathbf{k}_i} | \psi_a \rangle$  with  $D = D^{\text{def}} \ominus D^{\text{prist}}$  built from the defect and pristine atom lists; for  $n_{\text{pol}} = 2$  the local fold acts diagonally on the two spinor components while  $D$  carries the  $2 \times 2$  spin blocks of  $d_{\mu\nu}^{\text{so}}$  (spin-diagonal  $d_{\mu\nu}$  for scalar pseudopotentials). (ii)  $P_R$  is applied channel-wise, subtracting the active states of each channel. The active states are exact

eigenstates of the same Kohn–Sham  $H$ , so  $[H, P_A] = 0$  and the Krylov space below stays in  $\text{ran } P_R$  up to round-off; the projector is nevertheless re-applied with each re-orthogonalization as numerical hygiene. (iii) The sources are assembled and QR-factorized *in the bundle inner product*,

$$[\chi_1 \cdots \chi_{N_A}] = Q_1 R_0, \quad Q_1^\dagger Q_1 = \mathbb{1}_{N_A}, \quad (9)$$

with  $R_0$  ( $N_A \times N_A$ , upper triangular) stored: the sources enter the final resolvent only through it, and its singular values are those of  $V_{RA}$  (the basis of the rank compression discussed in Sec. VIII).

### B. Block recurrence

With  $Q_0 \equiv 0$ , steps  $j = 1, \dots, N_S$  perform

$$W = H Q_j, \quad (10)$$

$$A_j = Q_j^\dagger W \quad (\text{Hermitian; } \|A_j - A_j^\dagger\| \sim 10^{-14} \text{ monitored per step}),$$

$$W \leftarrow W - Q_j A_j - Q_{j-1} B_{j-1}^\dagger,$$

$$W \leftarrow W - \sum_{l=\max(1, j-m)}^j Q_l (Q_l^\dagger W) \quad (\text{window } m), \quad (11)$$

$$Q_{j+1} B_j = \text{qr}(W) \quad (B_j \text{ upper triangular}). \quad (12)$$

$H$  in (10) is the untouched Kohn–Sham Hamiltonian of the pristine primitive cell, applied channel-diagonally: kinetic term in  $\mathbf{G}$ -space, one FFT pair for the local potential, a projector ZGEMM for the nonlocal part.  $\Delta V$  never appears after the source step — the defect enters  $\Sigma^R$  only through  $R_0$ . The coefficient pair  $(A_j, B_j)$  is appended to the chain file; bundles are discarded as the window advances. Applying  $\Delta V$  (source step and the operator-identity gate below) is the cost driver: by Eq. (??) every output channel couples to every source channel, a triple loop over grid points, channels and columns.

Before the first step the code verifies the *operator identity*  $\langle \psi_b | \Delta V | \psi_a \rangle$  (recomputed by the fold machinery on probe columns) against the independently produced Born block  $M_{AA}$ ; agreement at the  $10^{-11}$  level is required and doubles as a cross-code closure when  $M_{AA}$  comes from EDI. Failures of this gate have caught every interface and convention bug encountered during development.

### C. Reorthogonalization: the rolling window

The classic Lanczos hazard — ghost eigenvalues from lost orthogonality — sets in only when a Ritz value converges (Paige). For the chains used here ( $N_S \leq 32$  blocks of width  $N_A$  in a space of dimension  $N_{\text{pw}}^\sigma N_k \sim 10^6\text{--}10^7$ ) no Ritz pair converges, and a window of  $m$  blocks in (11) suffices. This was verified rather than assumed: against

full reorthogonalization,  $m = 2$  reproduces the chain matrices to  $3 \times 10^{-14}$  on a scale of 31, the physical levels bit-for-bit, and the global orthogonality  $\max |Q^\dagger Q - \mathbb{1}| = 6.0 \times 10^{-9}$  identical to four digits;  $m = 1$  agrees with  $m = 2$  to 0.000000 meV. The window replaces the full block store ( $N_S + 1$  blocks) by  $m + 1$  slots reused modulo  $m + 1$ ; together with the in-flight work block this sets the memory floor of the method at  $(m + 2)$  bundles of width  $N_A$  (Sec. VIII).

### D. Continued fraction and evaluation

For any complex  $\omega$  the resolvent follows from the block continued fraction, evaluated from the bottom of the chain,

$$S_j = [\omega - A_j - B_j^\dagger S_{j+1} B_j]^{-1}, \quad \Sigma^R(\omega) = \frac{R_0^\dagger S_1(\omega) R_0}{N_k}. \quad (13)$$

The inverse is constructed, never expanded, so there is no convergence-radius condition: a dressed rest state that a Neumann ladder would fail to converge is carried exactly. One chain serves every  $\omega$  and every broadening.

Evaluation details. The recursion is initialized at the bottom,  $S_{N_S} = [\omega - A_{N_S}]^{-1}$ , and swept upward; each  $\omega$  costs  $N_S$  Hermitian solves of size  $N_A$  ( $\sim N_S N_A^3$  flops, seconds at  $N_A \sim 10^3$ ), independent of the plane-wave dimension — the chain file  $(R_0, \{A_j, B_j\})$  is the entire interface between the heavy solver and all post-processing. Static quantities (levels, fixed points) evaluate at real  $\omega$  between the Ritz poles; spectral functions evaluate at  $\omega + i\eta$  with the physical broadening. The truncation parameter is  $N_S$  alone: its converged value depends on where the poles of  $P_R H P_R$  sit relative to the evaluation window, which in turn depends on the active-window choice. The practical rule, established by a supercell-truth comparison at fixed  $N_S$ : keep every state with significant defect weight in the energies of interest *inside* the active manifold, so that  $\Sigma^R$  is smooth across the evaluation window. A manifold that demotes such states to the rest space (e.g., dropping the S- $p$ -dominated valence bands that carry the sulfur-vacancy deep levels) forces the rational approximant to build sharp pole structure with few terms; at  $N_S = 16$  this cost 150 meV on the deep-gap doublet and produced spurious in-gap Ritz features, while the wider manifold restored agreement with the supercell spectrum to 20–40 meV at identical  $N_S$ .

Two properties of Eq. (13) matter in practice. The poles of  $\Sigma^R$  are exactly the Ritz values of the block-tridiagonal Krylov Hamiltonian, so their number in any window can be counted by inertia (a block  $LDL^\dagger$  sweep) without computing eigenvectors. And the residue of a pole is  $R_0^\dagger |n\rangle \langle n| R_0 / N_k$ , which is rank one; a pole whose eigenvector has little weight on the source block contributes negligibly even if it lies inside the region of interest.

For the case studied, of the  $N_A N_S = 25\,344$  poles, 25

lie inside the active manifold and four inside the fundamental gap — yet a dense scan shows  $\|\Sigma^R\|$  rising smoothly and monotonically from 0.398 to 0.863 across the whole gap, so those four carry negligible residue. One pole near +2.65 eV does not:  $\|\Sigma^R\|$  reaches 41.9 and the diagonal changes sign. It is a dressed rest state about 1 eV above the conduction edge, and it is exactly the kind of object that defeats a truncated ladder.

### E. Frequency dependence in practice

Freezing  $\Sigma^R$  at a reference  $\omega_0$  is exact at  $\omega_0$  and degrades linearly away from it. In the case studied  $\omega_0$  lies below the gap, and freezing costs 18.2 meV on the  $a_1$  in-gap level and 23.5 meV on the deeper doublet — larger than every other error in the chain. It is also unnecessary. Because all poles of  $\Sigma^R$  lie outside the active manifold except those identified above,  $\Sigma^R$  is analytic across the gap with the nearest singularity more than 3 eV away, and a low-order Chebyshev fit on a handful of nodes reproduces a held-out node to 0.026 meV and the quasiparticle fixed points to 0.201 meV.

When a wider window or a very dense  $\omega$  grid is wanted, two remarks apply. The pair-Wannier transform of Sec. V is *linear*, so writing  $\tilde{V}(\omega) = \sum_i f_i(\omega) C_i$  places the entire frequency dependence on scalar coefficients: the  $C_i$  are transformed once and any  $\omega$  costs a weighted sum. Alternatively the pole-residue form is exact and, because the residues are rank one, its transform factorizes into single-index vector transforms,  $\mathcal{W}_n(\mathbf{R}_e, \mathbf{R}_p) = \tilde{u}_n(\mathbf{R}_e) \otimes \tilde{u}_n^*(\mathbf{R}_p)$ . In either case the continued fraction, not the transform, is the quantity to economize.

## V. WANNIER REPRESENTATION OF THE DRESSED VERTEX

### A. Pair-Wannier transform

Because the defect breaks translational symmetry,  $\tilde{V}(\mathbf{k}, \mathbf{k}')$  is not a function of  $\mathbf{k} - \mathbf{k}'$  and cannot be interpolated with a single index. We use the two-index construction of Ref. 9,

$$\mathcal{V}(\mathbf{R}_e, \mathbf{R}_p) = \frac{1}{N_k^2} \sum_{\mathbf{k}\mathbf{k}'} e^{i\mathbf{k}\cdot\mathbf{R}_e} e^{-i\mathbf{k}'\cdot\mathbf{R}_p} U^\dagger(\mathbf{k}) \tilde{V}(\mathbf{k}, \mathbf{k}') U(\mathbf{k}'), \quad (14)$$

with  $U(\mathbf{k})$  the Wannier gauge matrices. *Both* indices are electron Wannier lattice vectors:  $\mathcal{V}(\mathbf{R}_e, \mathbf{R}_p) = \langle w_{\mathbf{R}_e} | \Delta V | w_{\mathbf{R}_p} \rangle$ . The defect is not a third index; it is where  $\Delta V$  lives, hence the point about which the kernel decays. (An equivalent parametrization transforms the electron index at fixed momentum transfer and the transfer separately; the two are related by the shear  $\mathbf{R}'_e = \mathbf{R}_p - \mathbf{R}_e$ .)

The construction requires an isolated composite band group so that  $U$  is unitary; disentanglement would make  $P_A$  mesh-dependent. For MoS<sub>2</sub> the 11-band  $d + p$  manifold is such a group, giving  $\|U^\dagger U - \mathbb{1}\| = 2 \times 10^{-10}$  and a gauge-invariant spread  $\Omega_I = 17.70 \text{ \AA}^2$  with  $\Omega_D = 0.030 \text{ \AA}^2$ . Critically, the wannierization must be performed on the *same* wavefunctions used to build the vertex: independent diagonalizations differ by arbitrary per-state phases.

### B. The origin convention

$\mathcal{V}$  is built on Born–von–Kármán representatives, and images differ by  $N^{\text{sc}} \mathbf{a}$ . On the coarse mesh  $e^{i\mathbf{k}\cdot N^{\text{sc}} \mathbf{a}} = 1$ , so the choice of representative is invisible — the transform inverts exactly whatever is chosen. On a *finer* mesh it is not unity, and a poor choice aliases. When the defect sits at the box centre rather than the origin, choosing representatives about the origin piles the kernel onto the cell boundary and turns a 1 meV prediction into a 300 meV one. The conventional remedy is the translation

$$\Delta V(\mathbf{r} + \mathbf{R}_{\text{def}}) \iff M(\mathbf{k}, \mathbf{k}') \rightarrow e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{R}_{\text{def}}} M(\mathbf{k}, \mathbf{k}'), \quad (15)$$

after which the ordinary origin-centred Wigner–Seitz construction applies; choosing representatives about  $\mathbf{R}_{\text{def}}$  is the same operation.

$\mathbf{R}_{\text{def}}$  should be read from the structure — the atom present in the pristine cell and absent from the defect cell — rather than from the kernel maximum, which returns a representative valid only on the lattice it was found on:  $(3, 3) \equiv (-3, -3)$  modulo 6 but not modulo 12. Note the asymmetry: *truncating* a coarse-mesh kernel is insensitive to the choice; only interpolation is.

### C. Validation by leave-one-out

Locality of  $\mathcal{V}$  in both arguments is the assumption the whole route rests on, so it is measured rather than argued. From a direct  $144\text{-}k$  vertex in a single Wannier gauge we build the kernel from the  $\Gamma$  coset alone — a complete  $6 \times 6$  dataset, 6.2% of the pairs — and predict the rest. The kernel falls from 1.46 on site to  $2 \times 10^{-4}$  at  $6a$ , and the  $6 \times 6$  and  $12 \times 12$  curves, the same quantity measured in two Born–von–Kármán cells, coincide wherever they overlap: the smaller cell truncates early rather than aliasing.

Element-wise in the same gauge, and quoted as  $\delta M \text{ Ry} / N_k$  (the scale at which a vertex element enters  $H_{\text{eff}}$ ), the 93.8% of pairs never seen by the construction are predicted with median 0.744 meV (p90 1.03, max 2.44); the training pairs return at  $4 \times 10^{-15}$ , certifying the transform rather than the physics. Repeating the test on the *dressed* vertex at seven frequencies spanning the gap gives 0.61 meV at the bottom of the gap — better

than the bare vertex, since the extra decaying resolvent makes  $\tilde{V}$  more local — rising to 0.88 meV at the top.

The same test quantifies the cost of a smaller active manifold. A five-band model built from the highest valence and four lowest conduction bands (an isolated group, so no disentanglement) would make the vertex construction  $2.2\times$  and the reorthogonalization  $4.8\times$  cheaper, but interpolates  $4.4\times$  worse (median 3.30 meV, p90 8.62). This is not a gauge that needs more optimization: all projection choices give the same gauge-invariant  $\Omega_I = 21.81 \text{ \AA}^2$  against 17.70 for eleven bands, i.e. 4.36 versus  $1.61 \text{ \AA}^2$  per Wannier function. Excluding the anion  $p$  bands that hybridize with the manifold costs localization that no minimization recovers.

## VI. T-MATRIX AND SPECTRAL FUNCTION

### A. Real-space cluster

With  $G_A^0 = (\omega - \varepsilon_A)^{-1}$  the free active-space propagator,

$$T(\omega) = [1 - \tilde{V}(\omega)G_A^0(\omega)]^{-1}\tilde{V}(\omega), \quad G_A = G_A^0 + G_A^0 T G_A^0. \quad (16)$$

Solving Eq. (16) directly on a fine mesh is prohibitive. In the pair-Wannier representation it becomes

$$\mathcal{T} = \mathcal{V} + \mathcal{V} \mathcal{G}^A(\omega) \mathcal{T}, \quad (17)$$

which closes on a truncated real-space cluster because  $\mathcal{V}$  is short-ranged and, in the gap,  $\mathcal{G}^A$  decays exponentially. The dimension of Eq. (17) is set by the cluster radius, not by the fine mesh — this is the step that escapes the  $N_k^3$  wall of Sec. VIII. It is the Koster-Slater construction [3] with a first-principles, frequency-dependent vertex in place of a model potential.

### B. Host Green function

The fine mesh enters only through a Brillouin-zone sum,

$$\mathcal{G}^A(\delta\mathbf{R}; \omega) = \frac{1}{N_f^2} \sum_{\mathbf{k}_f} e^{i\mathbf{k}_f \cdot \delta\mathbf{R}} [(\omega + i\eta) - H^W(\mathbf{k}_f)]^{-1}, \quad (18)$$

with  $H^W(\mathbf{k}_f)$  Wannier-interpolated from the same wannierization that supplies  $U(\mathbf{k})$  in Eq. (14). Using the two objects from a single wannierization is a hard requirement:  $\mathcal{V}$  and  $\mathcal{G}^A$  must be expressed in the same orbital basis for their product to mean anything. As a consistency check, the interpolated host reproduces the valence-band maximum and the full active-manifold energy range of the direct calculation to all printed digits.

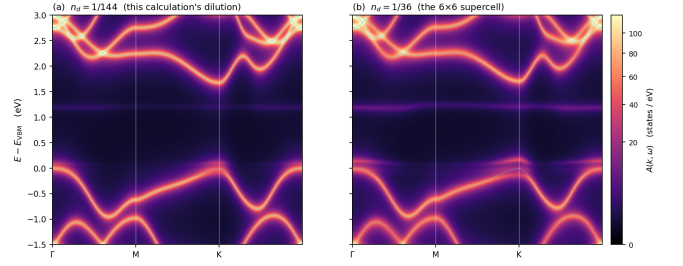


FIG. 1. Electron-defect spectral function  $A(\mathbf{k}, \omega)$  along  $\Gamma$ – $M$ – $K$ – $\Gamma$  for the relaxed sulfur vacancy, at two concentrations, with the bare interpolated bands overlaid. The flat in-gap lines are the Koster-Slater bound states; their energy is  $\mathbf{k}$ -independent by construction while their weight is not.

### C. Self-energy and spectral function

Transforming back,

$$T_{mn}(\mathbf{k}, \mathbf{k}'; \omega) = \sum_{\mathbf{R}_e \mathbf{R}_p} e^{-i\mathbf{k} \cdot \mathbf{R}_e} e^{+i\mathbf{k}' \cdot \mathbf{R}_p} \mathcal{T}_{mn}(\mathbf{R}_e, \mathbf{R}_p; \omega), \quad (19)$$

and Eqs. (1)–(2) follow. Two remarks. First, the diagonal  $T(\mathbf{k}, \mathbf{k})$  gives the spectral function, whereas transport requires the off-diagonal  $\sum_{\mathbf{k}'} |T(\mathbf{k}, \mathbf{k}')|^2 \delta(\varepsilon - \varepsilon_{\mathbf{k}'})$ ; both come from the same  $\mathcal{T}$ . Second, and central to the motivation,  $n_d$  in Eq. (1) is now a free physical parameter: the real-space cluster solves a genuine single-defect problem, so the concentration is no longer tied to the mesh as it is in Eq. (4).

In the gap  $\mathcal{G}^A$  is real, so bound states are the roots of  $\det[1 - \mathcal{V}\mathcal{G}^A(\omega)] = 0$  — a determinant containing no  $\mathbf{k}$ . Their energies are therefore  $\mathbf{k}$ -independent while their spectral weight varies along the path, the latter being the overlap of the localized state with each Bloch state.

### D. Application

Figure 2 shows the method applied unchanged to three point defects. The chains are indistinguishable in cost (16 blocks, 141 min, operator identity  $2 \times 10^{-11}$  to  $7 \times 10^{-12}$ ) because the cost depends only on the mesh and manifold dimensions, not on  $\Delta V$ . The physics is not: the vacancy drives  $\max |T| = 11.2$  and binds two in-gap states, whereas the isovalent substitutions reach only 0.44 and 0.17 and bind none. Since  $\mathcal{G}^A$  is real in the gap, the absence of a singularity in  $1 - \mathcal{V}\mathcal{G}^A$  is a statement, not a numerical near-miss.

## VII. VALIDATION

Each stage is gated separately, so that a failure is localized rather than absorbed into a final discrepancy. Table I collects the results.

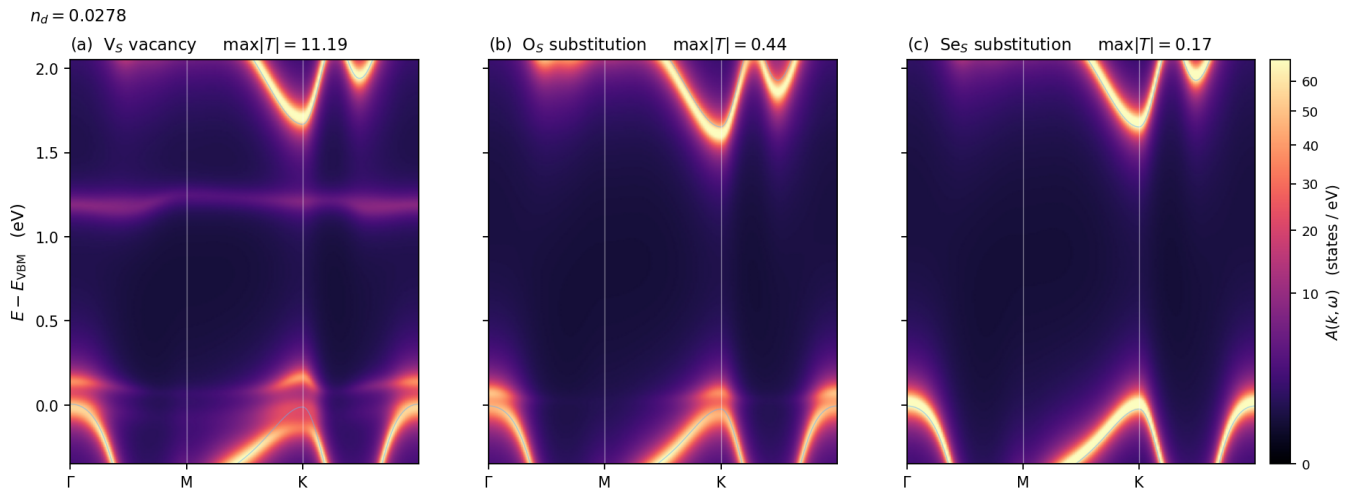


FIG. 2. The same pipeline applied to three point defects at  $n_d = 1/36$ : identical chains (16 blocks, 141 min), identical transform, identical cluster. A bound state requires  $1 - \mathcal{V}\mathcal{G}^A$  to become singular in the gap, where  $\mathcal{G}^A$  is real; only the vacancy achieves it. The isovalent substitutions leave the gap clean and merely broaden the host bands, strongly for O<sub>S</sub> and barely at all for Se<sub>S</sub>.

### A. Operator level

The application of  $\Delta V$  inside the Lanczos recursion, Eq. (??), is compared against the independently constructed vertex matrix by projecting the operator applied to known band states onto the band basis. The two paths share no code beyond the potential file. Agreement is  $2 \times 10^{-11}$  relative on the  $12 \times 12$  mesh and  $2 \times 10^{-14}$  on  $6 \times 6$ ; the test is run at the start of every production chain and is the gate for any change to the fold.

### B. Against a supercell reference

The coset construction of Sec. III C makes a supercell comparison possible at more than one supercell  $k$ -point while holding the concentration fixed. The three  $M$  cosets are symmetry-equivalent and the downfolding does not know it: run separately they return quasiparticle levels agreeing to every printed digit, matching the exact degeneracy of the supercell reference. A dedicated 107-atom calculation at the three  $M$  points supplies a reference no part of the downfolding has seen.

Levels agree with a rigid offset:  $+25.5 \pm 1.7$  meV at  $\Gamma$  (5/5 features) and  $+25.0 \pm 0.8$  meV at  $M$  (5/5). The two offsets agree to 0.5 meV — the constant is a property of  $\Delta V$ , not of  $\mathbf{k}$ . It is a potential-reference convention:  $\Delta V$  is vacuum-aligned while the reference is anchored on deep levels, and the difference is predicted with no fitting from two independently measured quantities ( $+17.9$  meV here), leaving residuals of  $+7.6 \pm 1.7$  and  $+7.1 \pm 0.8$  meV.

The same machinery bounds what is worth chasing. Running the finer mesh directly rather than by cosets dilutes the defect four-fold with no change of alignment (the same  $\Delta V$ , so the reference constant cancels exactly, and both meshes contain the same high-symmetry points

so the pristine edges are identical). The true in-gap levels move by  $-11.1$  and  $+3.7$  meV: the supercell reference itself carries 4–11 meV of periodic-image error, of the same size as the residual above.

### C. End to end

Finally the assembled chain — Wannier gauge, pair kernel, cluster truncation, fine-mesh  $\mathcal{G}^A$ , Dyson solve — is closed onto the coarse-mesh result. The deep in-gap bound state, a root of a determinant with no  $\mathbf{k}$  in it, comes out at  $+1.2019$  eV against  $+1.2023$  eV from direct diagonalization of  $H_{\text{eff}}(\omega)$ : 0.38 meV. A systematic error anywhere in the chain could not survive this comparison.

A second in-gap feature 68 meV above the valence edge differs by 7.3 meV, and doubling the frequency resolution moves it by only 1.5 meV, so it is not a grid artefact. At  $\eta = 50$  meV that state is hybridized with the valence continuum: it is a resonance, and the peak of  $|T|$  is not the same quantity as a quasiparticle fixed point computed on a discrete coarse spectrum that has no continuum to hybridize with.

## VIII. COMPUTATIONAL PARAMETERS, SCALING AND COST

### A. Parameters

Ground state and pristine band structure with QUANTUM ESPRESSO [10, 11], PBE [13],  $E_{\text{cut}}^{\text{wfc}} = 50$  Ry,  $E_{\text{cut}}^{\rho} = 200$  Ry, a 2D truncated Coulomb treatment, and a 24 Å vacuum. The defect supercell is  $6 \times 6$  (107–108 atoms), relaxed. The pristine calculation supplying the active manifold uses a  $27 \times 27 \times 216$  FFT grid, 20 bands

TABLE I. Validation summary. Energies in meV unless noted.

| Stage         | Test                                      | Result              |
|---------------|-------------------------------------------|---------------------|
| Fold          | $C_3$ degeneracy on commensurate grid     | 0–4 $\mu\text{eV}$  |
| Vertex        | operator identity, $12 \times 12$         | $2 \times 10^{-11}$ |
| Vertex        | operator identity, $6 \times 6$           | $2 \times 10^{-14}$ |
| Vertex        | direct vs. coset-diagonal blocks          | $2 \times 10^{-6}$  |
| Chain         | $N_S$ convergence ( $16 \rightarrow 24$ ) | 0.8                 |
| Downfold      | vs. supercell at $\Gamma$ (5/5)           | $+7.6 \pm 1.7$      |
| Downfold      | vs. supercell at $M$ (5/5)                | $+7.1 \pm 0.8$      |
| Reference     | periodic-image error of the reference     | 4–11                |
| $\omega$ fit  | held-out node                             | 0.026               |
| $\omega$ fit  | quasiparticle fixed points                | 0.201               |
| Interpolation | leave-one-out, bare vertex (median)       | 0.744               |
| Interpolation | leave-one-out, dressed vertex             | 0.61–0.88           |
| End to end    | cluster vs. direct, deep level            | 0.38                |

and a  $12 \times 12$   $k$ -mesh; the potential is band-limited to  $162 \times 162 \times 216$  per Sec. III. The active manifold is the 11-band  $d + p$  group,  $N_A = 1584$ . Chains use  $N_S = 16$  blocks at  $12 \times 12$  and  $N_S = 24$  at  $6 \times 6$ . Wannier functions from WANNIER90 [12], 11 per cell, Mo- $d$  and S- $p$  projections, no disentanglement. The  $T$ -matrix uses a cluster radius  $4a$  (61 cells, dimension 671), a  $96 \times 96$  mesh for Eq. (18), and  $\eta = 50$  meV.

Note that a host calculation must declare every atomic species appearing in any defect cell, including species absent from the pristine cell, since the nonlocal projector count is indexed by the *cell's* species list.

## B. Scaling

At fixed rank count the vertex application, Eq. (??), costs

$$t_{\text{fold}} \propto \frac{N_r n_b N_k^3}{n_{\text{pool}}}, \quad (20)$$

an  $N_k^2$  channel double loop times  $N_k$  columns per rank; full reorthogonalization carries the same  $N_k^3$  with an additional  $N_S^2$ . Measured,  $36 \rightarrow 144$   $k$ -points take the multiply from 4.1 to 261 s, i.e.  $64 \times = 4^3$ , confirming the

law. Memory scales as  $N_k^2$ , which is what limits the chain length. Reorthogonalization overtakes the vertex application around the ninth block.

Storing every block is what caps the chain length, so it is worth testing whether it is needed. Repeating an  $N_S = 24$  chain while orthogonalizing against only the last  $m$  blocks gives, for  $m = 2, 4, 8$  and for full: an orthogonality  $\max_{ij} \|Q_i^\dagger Q_j - \delta_{ij}\| = 6.04 \times 10^{-9}$  *identical to four figures*, bit-identical levels, chain matrices differing by  $3 \times 10^{-14}$  on a scale of 31, and zero Ritz values in the gap in every case. Full reorthogonalization does nothing here, for the reason the theory gives: loss of orthogonality is triggered by a converged Ritz value, and 24 blocks of width 396 in  $\sim 10^6$  dimensions converge none. (The residual  $\sqrt{\epsilon}$  is the single-pass classical Gram–Schmidt within a step, not a recurrence loss.) The gain is memory rather than speed —  $1.65\times$  faster, but  $Q_s$  falls from  $N_S + 1$  blocks to two, i.e.  $531 \rightarrow 62$  GB here and  $2.12 \text{ TB} \rightarrow 250 \text{ GB}$  with spin–orbit coupling, which moves an SOC  $12 \times 12$  calculation from eight nodes onto one. The scope is what was measured: a longer chain, or a spectrum that lets a Ritz value converge, would behave differently, so the implementation of record should be *partial* reorthogonalization — monitor the orthogonality, which is cheap, and reorthogonalize fully only when it degrades.

## C. Cost

On one 128-core node: the 144- $k$  chain takes 140 min (16 blocks, 261 s per block in the vertex application). Post-processing is dominated by the continued fraction, 6 s per frequency at  $N_A = 1584$ ,  $N_S = 16$ ; the cluster solve and the pair-Wannier transform are 0.5 s and 0.1 s per frequency. Frequencies are independent, and an interleaved eight-worker split delivers 450 frequencies in 6 min 25 s (0.85 s per frequency, a  $13.9\times$  speedup). The complete spectral function of Fig. 1 — 450 frequencies, 181  $k$ -points — therefore costs minutes once the chain exists, and any defect concentration afterwards is free because  $T$  is independent of  $n_d$ .

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